Harshit Poddar

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Seeking **graduate role** or **full-time** opportunity in Generative AI, Agentic AI, and Machine Learning. Specializing in state-space models, agentic LLM orchestration, computer vision, physical AI, and scientific machine learning. Open to relocation.

EDUCATION

KTH Royal Institute of Technology — M.Sc. in Machine Learning — Stockholm · Aug 2025 – Jun 2027 (expected)
Relevant Coursework: *Machine Learning Advanced (DD2434) · Deep Learning in Data Science (DD2424) · Deep Learning Advanced Course (DD2610) · Deep Generative Models & Synthesis (DD2601) · Artificial Intelligence & Multi Agent Systems (DD2438).*

Vellore Institute of Technology — B. Tech in Electronics & Communication Engineering — India · 2021 – 2025
CGPA: 4.4/5.0

EXPERIENCE

Massive Dynamic Sweden AB — AI Engineer (part-time) — Stockholm · Jul 2026 – Present

- Building the AI orchestration layer for *O-LISTIC*, an automated cybersecurity-policy platform; converting a manual pipeline into an end-to-end agentic workflow, **1 of 6 policy-generation stages migrated to tool-calling agents so far**.
- Exposing each migrated stage as a tool-calling agent behind typed FastAPI endpoints, with output validation, retry and fallback routing, and run tracing so every generated policy is reproducible and auditable.

STACK: Python · FastAPI · LLM APIs · agent orchestration · PostgreSQL · Git

Karolinska Institutet — Summer Intern — Stockholm · Jun 2026 – Present

- Generating **5,000+ synthetic** realistic fixational eye movement sequences by simulation, far more labelled sequences than lab measurement supplies and owning the simulation to training pipeline: parameter sweeps, labelling, versioning and reproducible configs.
- Modelling the resulting time series with a state-space architecture in PyTorch, benchmarked against diffusion, DiffWave and GAN baselines on distributional agreement with recorded data. Project details confidential.

STACK: Python · PyTorch · state-space models · Linux · Flow Matching

Tata Memorial Hospital (ACTREC) — Research Intern — India · Nov 2023 – Dec 2023

- Benchmarked U-Net against YOLOv8-L for nuclei segmentation on MoNuSeg (37 H&E slides, 7 organs): U-Net reached 0.82 recall and 0.63 IoU with stronger boundary delineation, while YOLOv8-L ran ~24% faster (61 ms vs 80 ms) at comparable precision. Applied Grad-CAM and SHAP for clinician review.

STACK: Python · U-Net · YOLO · Grad-CAM/SHAP · histopathology imaging

Earlier Internships — DRDO, ISRO, IIM Trichy — India · 2023 – 2024

- **DRDO** (Jun–Jul 2024) 8×60 phased-array model with a real-time MATLAB GUI and radar-degradation analysis · **ISRO** (Sep–Oct 2023) 16×16 phased array for MEOSAR signals in Ansys HFSS with MVDR/LCMV beamforming · **IIM Trichy** (Mar–Jun 2023) ETL and NLP sentiment pipeline over 5,000+ survey responses and 200+ TV advertisements.

TECHNICAL SKILLS

- **ML & deep learning** · PyTorch, state-space models, transformers, CNNs (U-Net, EfficientNet, YOLOv8/v11), explainable AI (Grad-CAM, SHAP), Hugging Face, Flow Matching
- **LLM systems & agents** · LangGraph, LangChain, RAG, tool-calling agents, Gemini APIs, vector search (pgvector, ChromaDB), structured-output validation
- **Engineering & MLOps** · Python, Docker, SQL, FastAPI, Streamlit, PostgreSQL/Supabase, Git, Linux, MLflow, GCP, PyQt5, MATLAB, LaTeX
- **Scientific & domain** · signal processing, Ansys HFSS, DICOM, FHIR, OMOP
- **Languages** · English (native) · Hindi (native) · Swedish (beginner)

PROJECTS

PatchPilot — Autonomous code healing agent [GitHub] — Jun 2026 – Present

- LangGraph state machine that runs a Python program, intercepts the traceback, and either returns a verified past fix from semantic memory (pgvector, similarity above 0.90) or asks Gemini 2.5 Flash for a patch — so an already-solved bug costs no LLM call.
- Human-in-the-loop gate before any file write, then re-execution and write-back of the verified fix as new memory; MLflow traces every prompt, token count and node latency.

STACK: Python · LangGraph · Gemini API · Supabase (PostgreSQL + pgvector) · MLflow · Streamlit

LungMate — AI-Driven Lung Cancer Diagnosis [GitHub] — Nov 2024 – Apr 2025

- Hybrid detector replacing the YOLOv11s backbone with EfficientNet-B3 + Swin Transformer + CBAM, trained on 6,789 annotated CT scans across four lung-cancer classes: **91.05% mAP@0.5, 90.12% precision, 84% accuracy** at 2.35 ms inference.
- Derived tumour diameter, area, perimeter, volume and mass from 2D bounding boxes via a calibrated pixel-to-cm factor, removing the need for 3D reconstruction; packaged in a PyQt5 GUI with 15 image-processing modes.

STACK: PyTorch · EfficientNet-B3 · Swin Transformer + CBAM · YOLOv11 · OpenCV · PyQt5 · DICOM

Embedded safety systems [GitHub] — Jan 2024 – Dec 2024

- High-altitude autonomous fall detection with sensor fusion and automated actuator deployment, and an ESP32 + LoRaWAN industrial safety wearable for hazardous-environment telemetry. One patent filed and one design patent granted.

STACK: ESP32 · LoRaWAN · sensor fusion · embedded C

PUBLICATIONS, PATENTS & AWARDS

Selected publications (full list on Google Scholar; author position in brackets):

- Poddar, H., Sivakumar, R. "LungMate: AI-driven lung cancer diagnosis with quantitative cancer assessment." IEEE SENNET 2025. **[1st]**
- Ramalakshmi, K., Rajagopal, S., Kulkarni, M. B., Poddar, H. "A hyperdimensional framework: RBP and GSN within CNNs for brain tumour classification." Biomedical Signal Processing and Control (Elsevier) 96(B), 2024. **[4th]**
- Thiagarajan, J. D. et al. (incl. Poddar, H.) "Analysis of banana plant health using machine learning techniques." Scientific Reports (Nature) 14:15041, 2024. **[7th]**

Patents · 1 design patent granted; 1 filed (IoT safety helmet). **Awards** · Raman Award (2×), VIT. **Other** · LLM evaluation participant, Google DeepMind and Andon Labs (paid honorarium) · **Member** · KTH AI Society and AESIR (KTH Rocketry).